

VALA RAHMATI

vrahmati@smu.edu ▪ <https://linkedin.com/in/vala-rahmati> ▪ 214-535-3501 ▪ Dallas, TX

EDUCATION

Ph.D., Operations Research, 2022 – Expected May 2027
Southern Methodist University (SMU), Dallas, TX, USA

GPA: 3.9/4.0

Advisor: Prof. Halit Üster

B.Sc., Industrial Engineering, 2017 – 2021
Amirkabir University of Technology, Tehran, Iran

GPA: 4.0/4.0

RESEARCH INTERESTS

- Transportation and Logistics
- Network Design
- Disaster Management
- Retail Management
- Large Scale Optimization
- Decomposition Algorithms

MANUSCRIPTS

Time-Dependent Integrated Network Design for Supply and Demand in Disaster Response June 2026

Vala R., & Halit Ü., *Transportation Science*, to be submitted in June 2026

Funding Source: National Science Foundation.

- Developed an **integrated, five-tier time-expanded network** that jointly **optimizes evacuation and relief distribution** by minimizing **operational costs and total evacuation–delivery times** while accounting for **congestion and temporal dynamics**.

Time-Phased Relief Supply Network Planning for Foreseen Disasters May 2025

Vala R., & Halit Ü., *Transportation Science*, published

Funding Source: National Science Foundation.

- Developed a **time-expanded network** model for large-scale disasters integrating **location, allocation, and distribution decisions** with **time-indexed variables** and **worst-case supply fulfillment** to enhance robustness and **tactical decision-making in relief logistics**.

PAPERS IN PROGRESS

Prediction of Evacuees' Route and Destination Selection in Time of Disasters May 2026 - Present

Vala R., & Halit Ü., Targeted Journal: *Transportation Science*,

Funding Source: National Science Foundation, Completion Rate: 50%

- Developed a **stochastic mixed-integer** model for a **5-tier, time-extended network** to **minimize total disaster relief and evacuation costs** while factoring in **road disruptions, donation/demand uncertainty, road capacity, and congestion constraints**.

Prediction of Evacuees' Route and Destination Selection in Time of Disasters March 2025 - Present

Vala R., Anna L., & Halit Ü., Targeted Journal: *Journal of Humanitarian Logistics and Supply Chain Management*,

Funding Source: National Science Foundation, Completion Rate: 40%

- **Working with an undergraduate student**, Anna Leo, to create a survey to gather information about **evacuees, previous experiences**, and their **decision making** in different disaster scenarios.
- Finding factors affecting these decisions to **predict destination and route selection of evacuees** in future disasters.

SCHOLARLY PRESENTATIONS

Time-Dependent Integrated Network Design for Supply and Demand in Disaster Response

- Paper Presentation, 2026 INFORMS Annual Meeting, San Francisco, California Oct 2026
- Paper Presentation, 2026 INFORMS TSL Conference, Massachusetts, Boston Jul 2026
- Paper Presentation, 2026 IISE Annual Conference, Arlington, Texas May 2026
- Poster Session, 2026 SMU Research Week, Dallas, Texas Mar 2026
- Paper Presentation, 2025 INFORMS Annual Meeting, Atlanta, Georgia Oct 2025

Time-Phased Relief Supply Network Planning for Foreseen Disasters

- Paper Presentation, 2025 IISE Annual Conference, Atlanta, Georgia May 2025
- Paper Presentation, INFORMS Student Chapter at Texas Tech University, Lubbock, Texas May 2025

- Poster Session, 2025 SMU Research Week, Dallas, Texas Mar 2025
- Paper Presentation, 2023 INFORMS Annual Meeting, Pheonix, Arizona Oct 2023
- Poster Session, 2023 SMU Research Week, Dallas, Texas Mar 2023

TEACHING

-
- OREM 3360 - Operations Research I**, Instructor (temporary assignment), SMU Sep 2023 – Oct 2023
- Delivered full lectures for few sessions while course instructor was away, prepared course materials, and supported student questions in class.
- OREM2360 - Engineering Economy**, Teaching Assistant, SMU Sep 2022 – May 2023
- Full responsibility in organizing materials and holding lab sessions.
- EMIS3360 - Operations Research I**, Teaching Assistant, SMU Jan 2022 – May 2022
- Sole instructor of lab classes to apply LP and MIP models in AMPL.
- Operations Research II**, Teaching Assistant, Amirkabir University of Technology Sep 2021 – Dec 2021
- 16 lab classes to develop MIP, NLP, and network design algorithms. Preparing lab materials.
- Operations Research I**, Teaching Assistant, Amirkabir University of Technology Jan 2021 – May 2021
- Sole instructor of lab classes to solve problems in LP models, duality, and applied related algorithms.
- Engineering Economy**, Teaching Assistant, Amirkabir University of Technology Jan 2021 – May 2021
- Developed 8 projects for related topics. Grading homework and exams.

PROFESSIONAL DEVELOPMENTS - TEACHING

-
- CLT Teaching Excellence and Networking**, INFORMS Annual Meeting Oct 2025, and Oct 2026
- A teaching workshop for data analytics, operations research, supply chain, and management science, with panels and collaborative sessions on job market preparation, teaching statements, curriculum design, ethical AI in teaching, inclusive pedagogy, mentorship, and networking.
- Center for Teaching Excellence**, SMU Sep 2024 – Dec 2024
- Participated in workshops on research based instructional design, effective pedagogy, and course assessment.

PROFESSIONAL DEVELOPMENTS - WORKSHOPS AND CERTIFICATIONS

-
- Moody School Leadership Academy**, SMU Sep 2025 – May 2026
- 12 sessions on teaching, research, and professional skills, culminating in a leadership portfolio and certificate.
- Future Faculty Fellow Program**, IISE Sep 2025 – May 2026
- A professional development cohort with monthly panels, and a culminating workshop at the IISE Conference.
- Faculty Colloquium**, 2025 IISE Annual Conference, Atlanta, Georgia May 2025
- NVIDIA Workshops**, May 2024
- Deep Learning, Accelerated Data Science, AI Applications for Anomaly Detection, Transformer Based Natural Language Process
- Introduction to ARC GIS**, GIS Research Support Workshops, SMU Jan 2024
- Ph.D. Students Colloquium**, 2023 INFORMS Annual Meeting, Pheonix, Arizona Oct 2023

MENTORING EXPERIENCE

-
- Graduate Mentor**, Operations Research & Engineering Management Department, SMU Jan 2025 - Present
- Helped an undergraduate student, Anna Leo, to create a survey and gather data about behavior of evacuees in terms of disasters and their decision-making process in different disaster scenarios.
 - Guiding the student in preparation and presentation of findings. The results will be submitted to *Journal of Humanitarian Logistics and Supply Chain Management*.

PROFESSIONAL EXPERIENCE

-
- Twin911Net: Digital Twin–Driven AI Agents for Network Resilience**, AT&T TwinNet 911 Hackathon Sep 2025
- Led the design of a **digital twin–driven AI co-pilot system** for 911 dispatchers, integrating **predictive modeling**, **network simulation**, and **LLM-based decision support** to enhance emergency communication resilience.
- Batch Optimization for E-Store Operations at Retail Stores**, Internship Project, H-E-B May 2025 – Aug 2025

- Independently scoped and led an **optimization-based project** on **batching and cubing processes** for **curbside operations under demand uncertainty**, collaborating with **business stakeholders** to define scope, validate assumptions, and deliver actionable solutions.

Flight Connection Delay Optimization, American Airlines Hackathon Feb 2025

- Directed development of a **predictive optimization framework** to **minimize crew layover** times between connecting flights using **real delay, route, and airport data**, and **presented results to airline analytics leaders**.

Travel Destination Prediction for Targeted Advertisement, American Airlines Hackathon Feb 2024

- Led an interdisciplinary team to develop a **data-driven travel prediction model** using **large-scale customer data**, achieving **first place among 15 competing teams across Texas**.

Freshmen Course Assignment Optimization, B.Sc. Thesis, Amirkabir University of Technology Nov 2021

- Developed a **mixed-integer optimization model** to **automate freshman course assignments**, improving scheduling efficiency and reducing conflicts in large academic networks.

HONORS AND AWARDS

Moody Dissertation Fellowship, Moody Graduate School, SMU Aug 2026 – May 207

2026 Outstanding Graduate Award, OREM Department, SMU May 2026

2025 Outstanding Graduate Award, OREM Department, SMU May 2025

1st Place in American Airlines Hackathon, American Airlines Mar 2024

Graduate Student Travel Grant, Moody Graduate School, SMU Oct 2025, May 2025, Oct 2023

1st Place in INFORMS PhD Student Chapter Video Contest, INFORMS Dec 2024

INFORMS Student Chapter Annual Reward (Honorable Mention), 2024 INFORMS Annual Meeting Aug 2024

TECHNICAL SKILLS

Programming Languages: Python, C++, R, SQL, MATLAB, C.

Software: Gams, AMPL, ArcGIS, Arena, SPSS, Microsoft – Excel, Word, PowerPoint.

Languages: English (Fluent), Persian (Native), Azerbaijani (Native), Turkish (Fluent), Arabic (Reading Proficiency).

SERVICES

CLT Teaching Excellence and Networking, 2026 INFORMS Annual Meeting, San Francisco, California Oct 2026
Session Chair

- Chaired a panel session on active learning strategies, showcasing how gamification, simulations, and case studies demystify complex quantitative concepts in analytics and operations management.

Moody Graduate School Webinar, Southern Methodist University, Dallas, TX, USA Oct 2025
Panelist

- Shared insights on doctoral funding opportunities and strategies for balancing academic responsibilities with graduate assistantships during a Moody Graduate School panel discussion.

Panelist

- Advised international students on graduate program requirements and academic expectations at SMU.

INFORMS Student Chapter, Southern Methodist University, Dallas, TX, USA Aug 2024 - Present
President

- Directed a 10-member executive board, successfully planning and executing over 10 high-impact annual events fostering innovation and knowledge growth in operations research and data analytics domains.

Vice President

Aug 2022 - Aug 2024

- Coordinated networking events connecting academic and industry professionals, featuring guest speakers.

REFERENCES

Dr. Halit Üster

Operations Research & Engineering Management,
Lyle School of Engineering,
Southern Methodist University,
Email: uster@smu.edu,
Relationship: Committee Chair.

Dr. Sila Çetinkaya

Operations Research & Engineering Management,
Lyle School of Engineering,
Southern Methodist University,
Email: sila@smu.edu,
Relationship: Committee Member.

Dr. R. Canan Savaskan-Ebert

Information Technology & Operations Management,
Cox School of Business,
Southern Methodist University,
Email: csavaskan@smu.edu,
Relationship: Committee Member.

Dr. Digvijay Boob

Operations Research & Engineering Management,
Lyle School of Engineering,
Southern Methodist University,
Email: dboob@smu.edu,
Relationship: Committee Member.